ANOMAX – AI Driven Anomaly Detection in Electricity Consumption

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***Abstract*-- Every day, smart meters and sub-meters in homes churn out mountains of electricity consumption data. If you dig into this data the right way, you can help everyone users, energy producers, and utility companies spot odd patterns and see where things are going sideways. That's where anomaly detection comes in. It's a cornerstone in machine learning, picking out weird blips and keeping small issues from turning into big ones, all while pushing for better energy efficiency. Here, we're talking about an AI-powered anomaly detection setup. It leans on machine learning algorithms like Isolation Forest and One-Class SVM, using Python to sift through consumption records. By comparing normal and abnormal usage over the same time periods, we saw clear spikes in electricity use during those oddball intervals. This helps drive sustainable energy habits, cuts down waste, and sharpens decision-making for today's power systems.**

Keywords—Electricity Consumption, Anomaly Detection, Sustainable energy, Artificial Intelligence, Machine Learning, Python.

I. INTRODUCTION

Global energy consumption remains heavily dependent on fossil fuels, with nearly 80% of electricity production still derived from non-renewable sources, contributing significantly to climate change [1]. Addressing this challenge requires both the adoption of sustainable energy solutions and the reduction of overall consumption. Smart meters and sub-meters installed in residential and commercial buildings generate vast amounts of electricity usage data daily. When analysed effectively, this data provides valuable insights into abnormal consumption patterns, enabling consumers and utility providers to identify inefficiencies and adopt energy-efficient practices [1].

Anomaly detection has emerged as a vital method in machine learning and data analysis, as it identifies deviations from standard consumption behaviour that may indicate energy wastage, equipment malfunction, or unauthorized usage [2]. Traditional monitoring systems merely report aggregate consumption, lacking the ability to highlight unusual patterns. In contrast, AI-driven anomaly detection frameworks leverage algorithms such as Isolation Forest and One-Class SVM to automatically distinguish between normal and anomalous consumption [2]. These models enhance decision-making and contribute to sustainable energy management by reducing waste and improving efficiency.

*Recent advancements in deep learning and transformer-based architectures have further strengthened anomaly detection capabilities, offering improved accuracy, scalability, and adaptability to large-scale electricity datasets [3]. By integrating these approaches, AI-based systems can provide robust, real-time detection of anomalies, supporting both individual consumers and utility companies in their efforts to promote sustainability. The proposed project, **ANOMAX**, aims to design and validate such a framework, demonstrating its effectiveness in identifying abnormal electricity consumption patterns and contributing to global climate change mitigation efforts [1][2][3].*

II. PROBLEM STATEMENT & HYPOTHESIS

A. ***Problem statement**: Most electricity monitors just show you how much power you use, and that's it. They don't notice when something's off or help you figure out where the issue is. This project flips that around by using machine learning to catch odd electricity patterns in real time.*

B. ***Hypothesis**:*

- Hypothesis 1: Unsupervised machine learning methods like Isolation Forest and One-Class SVM can actually spot strange electricity usage—like sudden jumps or drops—pretty well.
- Hypothesis 2: If we automate data cleaning and team it up with machine learning, people save time and catch unusual patterns faster and more accurately than with old-school methods.
- Hypothesis 3: Breaking down electricity data by sub-meters doesn't just help pinpoint where weird usage starts—it also gives a much clearer picture of how the whole system's running.

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III. LITRETURE SURVEY

Artificial intelligence has emerged as a vital instrument for spotting irregularities in power usage. A review by Himeur et al. (2021) highlighted that machine-learning, deep-learning, and hybrid AI techniques are capable of identifying such anomalies, although issues related to scalability and real-time performance persist [1]. Kardi et al. (2021) showed that deep-learning architectures can learn both temporal and spatial consumption patterns, rendering them appropriate for extensive monitoring applications [2]. More recently, Mu et al. (2023) presented transformer-based models that are trained with artificially generated anomalies, achieving superior accuracy and robustness [3]. In a complementary line of work, Zhang et al. (2020) introduced an unsupervised anomaly-detection framework grounded in feature engineering [4]. Their method focused on building an optimal feature set through idea-generation sessions, variance-driven selection, and an enhanced minimum-Redundancy-Maximum-Relevance (m-RMR) algorithm that incorporates the maximum information coefficient (MIC), followed by density-based clustering. Tests on load data from an industrial park demonstrated that this approach yields higher detection accuracy and better generalization than conventional techniques.

TABLE NO.1. LITRETURE SURVEY

Serial. No.	Literature survey table		
	Title	***year***	***Summary***
1.	*Artificial intelligence based anomaly detection of energy consumption in buildings – Y. Himeur et al.*	*2021*	*ML, DL, and hybrid AI models detect anomalies; ML is simple but limited, DL handles complex patterns but needs big data, hybrids balance both. Challenges include scalability and real-time detection. Hybrid approaches can improve accuracy.[1]*
2.	*Anomaly Detection in Electricity Consumption Data using Deep Learning – M. Kardi et al.*	*2021*	*DL models capture temporal and spatial consumption patterns, making them suitable for large-scale monitoring. Applying DL helps identify anomalies effectively.[2]*
3.	*Detecting anomalous electricity consumption with transformer and synthesized anomalies – T. Mu et al.*	*2023*	*Transformer-based models trained with synthetic anomalies improve robustness, accuracy, and adaptability. They enhance scalability in anomaly detection frameworks [3].*
4.	*Unsupervised Detection of Abnormal Electricity Consumption Behavior Based on Feature Engineering – W. Zhang et al.*	*2020*	*Proposed unsupervised detection using brainstorming-based feature construction, variance selection, and improved m-RMR with MIC, followed by density clustering. Tests on industrial park load data showed higher accuracy and generalization. Feature engineering combined with clustering improves anomaly detection.[4]*

IV. EASE OF USE

The use of AI-driven Anomaly Detection in electricity consumption machine is an vital aspect that complements its usability for a extensive variety of customers, inclusive of residential clients, researchers, and utility carriers. The system is designed to be easy, green, and on hand without requiring superior technical expertise .The following features make contributions to the usability of the device:

- *Person-pleasant Interface: The system affords an intuitive interface through tools which includes jupyter notebook or dashboards, enabling users to without problems upload and analyse energy consumption information.*
- *Automatic Processing: The proposed device mechanically plays information preprocessing, version schooling, and anomaly detection the use of device mastering algorithms like Isolation wooded area and One-class SVM.*
- *minimal person Intervention: Customers are most effective required to offer the dataset, and then the machine executes all operations independently, reducing guide attempt and complexity.*
- *Visualization of results: The system offers output using graphical representations such as time-series plots, making it less complicated to identify unusual consumption patterns.*
- *Cost-effective Implementation: The usage of open-source tools which include Python, Pandas, NumPy, and Scikit-research ensures that the system is lower priced and widely accessible.*
- *Scalability and flexibility: The system may be applied to distinct datasets and can be prolonged to real-time applications with minimum changes.*

Overall, the proposed system ensures a balance between technical efficiency and user convenience, making it suitable for practical deployment in energy monitoring applications.

V. METHEDOLOGY

This method breaks down into a few main steps: collecting data, cleaning it up, shaping features, building models, spotting anomalies, and, finally, visualizing the results. The setup works with both smart-meter and sub-metering data—so you don't just catch when something looks off, you can track down where the wasted electricity might be coming from [1], [4].

First, you start by gathering electricity usage data from reliable sources. It all comes as time-series info, usually in CSV files. These hold a history of power usage over different periods. Sub-metering data gives you a closer look, almost like zooming in to see which devices or parts of a building are eating up electricity, which makes it a lot easier to catch what's causing spikes [1].

Next, there's preprocessing. This is a fancy way of saying you clean the data. That means dealing with missing numbers, stripping out noise and errors, normalizing the values, and turning timestamps into something useful. Once this is done, the dataset's nice and clean—ready for the models to get to work [4].

After that comes feature engineering. Here, you pull out the key bits: time intervals, actual consumption numbers, and all those detailed sub-meter readings. These are what help the system learn what "normal" power use looks like and spot when things start wandering off track [4].

The core of the system is the modeling. It relies on unsupervised machine learning algorithms to detect odd patterns:

- *Isolation Forest splits up data points randomly to single out the weird ones-anomalies pop up fast because they're different [2].*
- *One-Class SVM draws a border around what's "normal," so when something steps outside, the model flags it [2], [3].*

Once you train these models on historical data, they have a solid grip on what typical usage patterns look like.
Then comes the real action-anomaly detection. The trained models scan through the data and flag points as either normal or out of the ordinary. These anomalies usually show up as sudden surges or big drops in energy use-things you'd want to notice right away because they often point to waste or something else unusual [2], [3].
For visualization, everything gets plotted on charts, usually time-series graphs. Anomalies stand out, so users can quickly see where the odd patterns are coming from and get a clear view of when and how electricity is being used [3].
Finally, there's evaluation. The system gets judged by how accurately it catches these unusual patterns. The flagged anomalies get compared with normal activity from the same time frame to see if the system really knows what it's doing. This way, you know if it's actually helping spot wastage and not just raising false alarms [2], [3].

VI. RESULT & DISCUSSION

We built and tested the system on real electricity consumption data, and it worked well. It spotted spikes and sudden drops in power use without missing a beat [2], [3]. The Isolation Forest algorithm handled anomaly detection quickly and efficiently, and One-Class SVM picked up on patterns that strayed from the norm [2], [3]. Using sub-metering data made it easier to pinpoint exactly where electricity was being wasted [1]. The visualizations made it simple to see when and where odd usage happened-the anomalies stood out right away [3].

In the end, the system achieved accurate detection, fast performance, and demonstrated suitability for real-world energy monitoring applications [1], [2], [3].

VII. SCOPE AND APPLICATIONS

We built and tested the ANOMAX system on real electricity consumption data, and it worked well. It spotted spikes and sudden drops in power use without missing a beat. The Isolation Forest algorithm handled anomaly detection quickly and efficiently, and One-Class SVM picked up on patterns that strayed from the norm. Using sub-metering data made it easier to pinpoint exactly where electricity was being wasted. The visualizations made it simple to see when and where odd usage happened—the anomalies stood out right away. In the end, the system nailed accurate detection, worked fast, and felt ready for real-world energy monitoring.

VIII. CONCLUSION

This study dives into how machine learning can catch unusual electricity usage like sudden spikes or unexpected drops—without needing labelled data. The algorithms Isolation Forest and One-Class SVM effectively perform anomaly detection, as supported by recent research [2], [3]. Bringing in sub-metering data makes it easier to identify where the electricity wastage might be occurring [1].

The system stands out because it is reliable and cost-effective. It works well with smart-meter data, making it suitable for residential and small-scale applications. Overall, this approach helps improve energy efficiency and supports better decision-making in electricity usage [1], [2], [3].

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